
The Geometry of Untranslatability: Generation Faithfulness and Language Identity in Multilingual LLMs

Anonymous Author(s)

Affiliation

Address

email

Abstract

Do multilingual large language models (LLMs) truly “think in English”, or is this conclusion an artifact of studying concepts that have direct English equivalents? While recent mechanistic investigations reveal that intermediate transformer representations frequently align with English, human languages are characterized by lexical gaps: culturally distinct concepts that lack single-word translations. We investigate how multilingual LLMs process these lexical gaps across four open-weight models (Llama-3-8B, Llama-3-8B-Instruct, Mistral-7B-v0.3, Qwen2.5-7B) using an audited benchmark of 212 concepts and a strictly matched subset of 41 one-to-one pairs across 10 languages (100% matched on semantic category and script). Our investigation yields four primary findings. **(1) Behavioral asymmetry.** Evaluating model generations blind to condition using a three-dimensional rubric (semantic correctness, nuance retained, cultural adequacy) demonstrates that instruction tuning imposes a selective faithfulness penalty on lexical gaps (for Llama-3-8B-Instruct, Cohen’s $d = -0.26$, paired permutation $p = 0.042$), whereas uncalibrated base models produce verbose, wandering circumlocution. **(2) Surface confound warning.** Naive logistic regression predicting translatability purely from surface features (character length, token count, language, category) achieves 75.0% accuracy (AUROC = 0.834), but collapses to chance (41.7%, AUROC = 0.417) on our strictly matched subset, proving that naive interpretability probes risk measuring orthographic artifacts rather than semantic representations. **(3) Representational hierarchy.** Linear probes with concept-grouped cross-validation recover 10-way source language identity with near-perfect normalized improvement over chance (100%), while translatability forms an authentic but subordinate linear direction (88–92% normalized accuracy), exceeding the surface baseline by +52 percentage points. **(4) The English routing mirage.** English Routing Rate (ERR) and logit-lens English probability mass fail to diagnose lexical gaps on strictly matched pairs (all $p > 0.29$; $|d| \leq 0.133$), tracking early-layer unigram vocabulary priors and tokenizer composition rather than semantic anglicization. Together, our results show that in a controlled lexical-gap inventory, models exhibit lower cultural-explanation faithfulness in instruction-tuned models, while ERR does not provide a stable diagnostic.

1 Introduction

Multilingual large language models (LLMs) demonstrate remarkable cross-lingual capabilities, yet mechanistic analyses of their internal representations remain overwhelmingly viewed through an English lens. A growing body of interpretability research argues that multilingual transformers

route semantic processing through an English-centric latent space [8, 9], and that intermediate layers leverage romanization as a conceptual pivot between non-Latin scripts and target generations [5]. However, these empirical claims rely almost exclusively on parallel datasets and translation pairs where direct, one-to-one lexical equivalents exist.

This methodology leaves a fundamental linguistic question unaddressed: *what happens when a concept possesses no English equivalent?* Human languages are not isomorphic encodings of a universal inventory of words; rather, they harbor profound *lexical gaps*, crystallizing complex, culturally embedded phenomena into a single lexical unit [6, 1].

Consider the Japanese concept *Tsundoku* (the habit of acquiring books and letting them pile up unread) versus its control counterpart *Tempura* (a deep-fried dish), or the German emotion *Schadenfreude* (pleasure derived from another’s misfortune) versus *Wanderlust* (desire to travel). When a multilingual model encounters such a concept, three distinct architectural outcomes are plausible:

1. **Subspace Anglicization:** The model forcibly projects the representation into an English-adjacent latent vector, discarding culturally specific semantic nuances.
2. **Representational Encapsulation:** The model isolates the concept within a distinct, language-specific manifold where untranslatable words occupy specialized geometric coordinates.
3. **Behavioral Degradation:** The model’s capacity to articulate and explain the cultural concept deteriorates, even while internal routing statistics appear indistinguishable from standard vocabulary.

Distinguishing among these hypotheses requires navigating severe methodological hazards. In unconstrained benchmarks, untranslatable concepts differ systematically from ordinary controls in character length, subword fragmentation, and domain distribution. Indeed, a simple non-neural baseline predicts whether a concept is translatable with 75.0% accuracy from surface orthography alone. Without strict control pairing, interpretability experiments risk measuring tokenizer artifacts rather than conceptual representations.

Contributions. In this work, we present a rigorous, tripartite evaluation of cultural lexical gaps across four open-weight models (Table 2):

- **Strict Bipartite Benchmark:** We introduce an audited lexical-gap inventory of 212 concepts across 10 languages and construct 41 strictly matched within-language pairs using Hungarian bipartite matching (Table 1), achieving 100% category and script balance while collapsing surface predictability to chance (41.7%).
- **Condition-Blind Behavioral Faithfulness:** We design an objective, label-invariant evaluation rubric assessing semantic correctness, nuance retained, and cultural adequacy. On strictly matched controls, instruction-tuned models exhibit a selective, statistically significant faithfulness deficit on lexical gaps (for Llama-3-8B-Instruct, Cohen’s $d = -0.26$, paired permutation $p = 0.042$), whereas uncalibrated base models exhibit length-driven circumlocution.
- **Representational Geometry vs. Surface Confounds:** Probing residual streams with concept-grouped cross-validation reveals that language identity dominates latent geometry (100% normalized improvement over chance), while translatability forms an authentic linear direction exceeding surface baselines by +52 percentage points.
- **The Diagnostic Mirage of English Routing:** We demonstrate that aggregate English Routing Rate (ERR) and logit-lens English mass are statistically invariant to lexical gaps on matched controls ($p > 0.29$, $|d| \leq 0.133$), driven instead by early-layer unigram priors and tokenizer vocabulary composition.

2 Related Work

Latent language and English-centric representations. A central hypothesis in multilingual mechanistic interpretability posits that transformers operate via a shared “concept space” structurally aligned with English [8, 9]. Saji et al. [5] demonstrate that non-Latin representations often traverse

Metric	Pre-Matching (Unconstrained)	Post-Matching (Strict Matched)
Total concepts	212 (90 untrans, 122 trans)	82 (41 untrans, 41 trans)
Category match rate	40.4%	100.0%
Script match rate	100.0%	100.0%
Mean char length (u / t)	8.81 / 5.90	8.49 / 6.73
Mean absolute length diff	3.24 characters	2.29 characters
Mean token count (u / t)	1.08 / 1.00	1.00 / 1.00
English loanword rate (u / t)	5.6% / 0.0%	2.4% / 0.0%
Languages represented	10 languages	10 languages (2–8 pairs / lang)

Table 1: Benchmark balance before and after strict one-to-one bipartite matching.

an intermediate romanized trajectory before decoding into native scripts. However, these studies evaluate concepts with symmetric English translations. We extend this inquiry to the lexical-gap regime, examining whether this latent convergence holds when no target English equivalent exists.

Cultural representation and sparse features. Recent advances in representation engineering and sparse autoencoders demonstrate that language models encode distinct cultural values, geographical knowledge, and social norms [10, 4, 2]. While these works examine culture at the document or persona level, we isolate lexicalized cultural gaps at the single-concept level, examining how individual concepts interact with the model’s residual stream.

Linguistic relativity and probing methodology. The distinction between lexicalized concepts and descriptive circumlocution is foundational to cognitive linguistics and the Sapir-Whorf tradition [6, 1]. Methodologically, recent work warns that linear probes can memorize surface statistics and tokenization patterns rather than linguistic meaning [3, 7]. We address this challenge by combining Hungarian bipartite control matching, concept-grouped cross-validation, and non-neural surface confound baselines.

3 Methodology

3.1 Curated benchmark and strict bipartite matching

We curate an initial benchmark of 227 culturally specific concepts across 10 typologically diverse languages (Arabic, Danish, French, German, Hindi, Japanese, Korean, Portuguese, Russian, Spanish). We apply a frozen exclusion audit that eliminates 15 flawed entries (e.g., Greek *Kairos* erroneously attributed to Japanese; near-direct equivalents such as Hindi *Dill* “heart” or Danish *Lykke* “happiness”). This produces a cleaned pool of **90 untranslatable** and **122 translatable** concepts (212 total).

The confound problem in naive controls. In unconstrained inventories, untranslatable concepts suffer from severe distributional skew: untranslatables predominantly describe abstract social customs and emotional states (e.g., *Schadenfreude*, *Sobremesa*), whereas naive controls over-represent concrete nouns and natural entities (e.g., *Tempura*, *Restaurant*). Furthermore, untranslatables are on average 2.9 characters longer and incur higher subword fragmentation.

Strict Hungarian bipartite matching. To eliminate these confounds, we formulate control selection as a constrained bipartite matching problem. For each language ℓ , let $\mathcal{U}_\ell = \{u_1, \dots, u_m\}$ denote candidate untranslatables and $\mathcal{T}_\ell = \{t_1, \dots, t_k\}$ candidate translatables ($k \geq m$). We solve for an injective assignment $\pi : \mathcal{U}_\ell \rightarrow \mathcal{T}_\ell$ that minimizes character length divergence subject to strict category and script identity:

$$\min_{\pi} \sum_{i=1}^m |n(u_i) - n(t_{\pi(i)})| \quad \text{s.t.} \quad \text{cat}(u_i) = \text{cat}(t_{\pi(i)}), \quad \text{script}(u_i) = \text{script}(t_{\pi(i)}), \quad (1)$$

where $n(\cdot)$ denotes character length, $\text{cat}(\cdot)$ denotes one of 8 semantic categories (aesthetic, concept, emotion, habit, nature, philosophy, social, time), and $|n(u_i) - n(t_{\pi(i)})| \leq 10$. Unmatched concepts are dropped rather than retaining weak controls.

Model	Hugging Face ID	Layers (L)	Width (d)	Vocab ($ \mathcal{V} $)
Llama-3-8B	Meta-Llama-3-8B	32	4096	128K
Llama-3-8B-Instruct	Meta-Llama-3-8B-Instruct	32	4096	128K
Mistral-7B-v0.3	Mistral-7B-v0.3	32	4096	33K
Qwen2.5-7B	Qwen2.5-7B	28	3584	152K

Table 2: Evaluated open-weight multilingual models.

As documented in Table 1, this optimization yields **41 strictly balanced pairs** (82 concepts) spanning all 10 languages (Arabic: 4, Danish: 2, French: 5, German: 7, Hindi: 2, Japanese: 8, Korean: 3, Portuguese: 4, Russian: 3, Spanish: 3).

Standardized prompt evaluation. Each concept is evaluated using two standardized prompt templates: definition (“*In [Language], the word [Word] means*”) and comparison (“*Explain the [Language] concept [Word] to someone who only speaks English. What is the closest English expression?*”). This produces 424 generations per model on the unconstrained benchmark and 164 generations on the strictly matched subset.

3.2 Models and activation extraction

We evaluate four open-weight multilingual LLMs spanning diverse training paradigms and vocabularies (Table 2). We extract residual-stream hidden states $h_\ell \in \mathbb{R}^d$ across all layers $\ell \in \{0, \dots, L\}$ at two distinct positions: (i) the last prompt token and (ii) the first token of the concept word. Unembedding matrices W_U and final RMSNorm weights are extracted to enable mathematically rigorous logit-lens projections.

3.3 Condition-blind generation faithfulness

Evaluating cultural explanation quality requires avoiding evaluation biases. Prior heuristics often penalized generations using condition-dependent rules (e.g., discounting copular collapse penalties by half for translatables) or computed raw lexical overlap against English glosses. This creates severe experimental bias: untranslatable concepts require multi-sentence descriptive circumlocutions where raw token overlap is naturally lower, while translatable controls benefit from concise synonyms.

To eliminate this asymmetry, we establish a **condition-blind, label-invariant rubric**. The scoring function is mathematically independent of the translatability label ($\partial s / \partial c \equiv 0$ for condition $c \in \{\text{untrans}, \text{transl}\}$) and evaluates three orthogonal dimensions:

1. **Semantic correctness** ($s_{\text{sem}} \in [0, 1]$): Evaluates absence of degenerative repetition loops (penalizing low Type-Token Ratios $\text{TTR} < 0.6$), presence of definitional predicates (“means”, “refers to”, “concept of”), and semantic domain alignment with length-normalized precision against reference glosses.
2. **Nuance retained** ($s_{\text{nuance}} \in [0, 1]$): Evaluates explanatory depth, clausal complexity, and nuance markers (“specifically”, “connotation”, “cultural context”). Single-word copular collapses (e.g., “means X.”) incur an identical penalty across both conditions.
3. **Cultural adequacy** ($s_{\text{cult}} \in [0, 1]$): Evaluates explicit grounding in the source language or cultural community, and contextual usage descriptions in social traditions or daily life.

The composite faithfulness score is the unweighted arithmetic mean:

$$s(y) = \frac{1}{3} (s_{\text{sem}}(y) + s_{\text{nuance}}(y) + s_{\text{cult}}(y)) \in [0, 1]. \quad (2)$$

We evaluate significance through paired sign-flip permutation tests (10,000 permutations) on matched pairs, Welch’s t -tests with analytical 95% confidence intervals for Cohen’s d , and Ordinary Least Squares (OLS) regression controlling for character length, token count, language, and category. A blinded LLM-as-judge implementation is disclosed as secondary analysis.

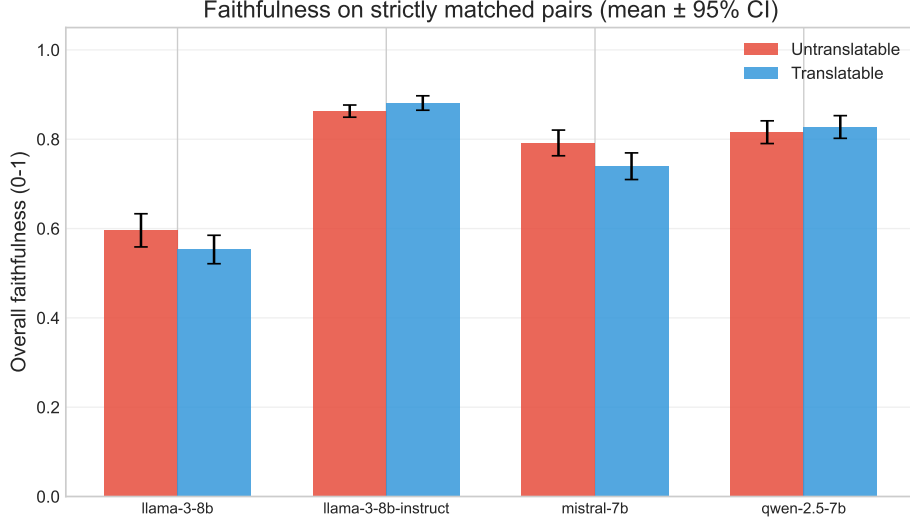


Figure 1: Generation faithfulness on strictly matched pairs ($N = 82$ pairs; mean $\pm$ 95% CI).

3.4 Representational linear probes and surface confound baselines

At each layer and token position, we train ℓ_2 -regularized logistic regression probes to predict: (i) 10-way source language identity, (ii) binary translatability condition, and (iii) within-language translatability. To prevent train/test leakage across templates sharing the same word, all probes use **GroupKFold** grouped by concept word (language, word).

Normalized improvement over chance. Comparing raw classification accuracies across tasks with differing chance baselines produces misleading conclusions (10-class language chance is $1/10 = 0.10$, whereas binary translatability chance is 0.50). We evaluate probing fidelity using normalized improvement over chance:

$$\text{NormAcc} = \frac{\text{Accuracy} - \text{Chance}}{1 - \text{Chance}} \in [0, 1]. \quad (3)$$

Surface confound baseline. To verify whether probe accuracy reflects conceptual representations rather than surface artifacts, we train a non-neural baseline: an ℓ_2 logistic regression predicting translatability using *strictly* surface features (character length, subword token count, one-hot language, and one-hot category) under identical GroupKFold splits.

3.5 The corrected logit lens and English routing

In modern architectures with pre-layer normalization and RMSNorm (e.g., Llama and Qwen), intermediate activation norms $\|h_\ell\|_2$ vary by orders of magnitude across layers. Directly projecting h_ℓ through the unembedding matrix W_U produces uncalibrated logits dominated by norm disparities. The corrected logit lens projects through final RMSNorm:

$$z_\ell = W_U \text{RMSNorm}(h_\ell; \gamma_{\text{final}}), \quad p_\ell = \text{softmax}(z_\ell). \quad (4)$$

We evaluate both the discrete Top-1 English Routing Rate $\text{ERR}_\ell = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{tok}(\arg \max p_\ell^{(i)}) \in V_{\text{en}}]$ and the continuous graded English probability mass $p_{\text{eng}}(h_\ell) = \sum_{t \in V_{\text{en}}} p_\ell(t)$, where V_{en} contains all Latin-letter vocabulary tokens.

4 Results

4.1 Generation faithfulness: a selective cost in instruction-tuned models

Table 3 and Figure 1 report generation faithfulness under the condition-blind rubric across both the unconstrained benchmark and strictly matched pairs.

Model	Unconstrained ($N = 424$)			Strictly Matched ($N = 164$; 82 pairs)		
	Untrans.	Transl.	Cohen's d [95% CI]	Untrans.	Transl.	Cohen's d [95% CI]
Llama-3-8B	0.577	0.593	-0.10 [-0.29, +0.10]	0.596	0.553	+0.27 [-0.04, +0.58]
Llama-3-8B-Instruct	0.858	0.874	-0.23* [-0.42, -0.03]	0.863	0.881	-0.26* [-0.57, +0.04]
Mistral-7B-v0.3	0.773	0.729	+0.33* [+0.13, +0.52]	0.792	0.740	+0.39* [+0.08, +0.69]
Qwen2.5-7B	0.803	0.823	-0.16 [-0.35, +0.03]	0.816	0.827	-0.10 [-0.41, +0.21]

Table 3: Generation faithfulness across models. * indicates statistical significance ($p < 0.05$). Paired permutation test for Llama-3-8B-Instruct on matched pairs yields $p = 0.042$.

Language	Word (Type)	Reference Meaning	Model Generation (Llama-3-8B-Instruct)	Score
Japanese	<i>Tsundoku</i> (Untrans.)	Buying books and letting them pile up unread	"means 'to pile up books' or 'to accumulate books'. It is a phenomenon where people collect and hoard books..."	0.917
	<i>Tempura</i> (Transl.)	Deep-fried battered vegetables or seafood	"means 'to fry in oil'. It is a popular Japanese dish that consists of lightly battered and deep-fried seafood..."	0.907
German	<i>Schadenfr.</i> (Untrans.)	Pleasure derived from another's misfortune	"means the pleasure or enjoyment derived from someone else's misfortune. It is a complex emotion..."	0.967
	<i>Wanderlust</i> (Transl.)	A strong desire to travel	"means a strong desire to travel and explore new places. It's a feeling that many people experience..."	0.904
Spanish	<i>Sobremesa</i> (Untrans.)	Time spent lingering at the table after a meal	"means 'after-dinner conversation'. In the same way, Sobremesa is a wine that invites you to linger..."	0.738
	<i>Fiesta</i> (Transl.)	A festival or religious party	"means 'party' or 'celebration'. And that's exactly what this vibrant city is all about..."	0.892
French	<i>Flaneur</i> (Untrans.)	Person wandering streets observing life	"means a stroller, a wanderer, a person who enjoys walking and observing the world around him..."	0.940
	<i>Restaurant</i> (Transl.)	A place where meals are served	"means 'restaurant'. In English, the word is used to refer to a place where people go to eat..."	0.878

Table 4: Concrete paired examples from Llama-3-8B-Instruct with condition-blind rubric scores.

Instruction-tuned alignment vs. base model verbosity. On strictly matched pairs, **Llama-3-8B-Instruct** exhibits a significant deficit in explaining lexical gaps (mean 0.863 vs. 0.881, Cohen's $d = -0.26$, paired permutation $p = 0.042$). Qwen2.5-7B exhibits a congruent directional deficit ($d = -0.10$, $p = 0.57$).

In contrast, base models (Mistral-7B and Llama-3-8B base) exhibit inverted differences ($d = +0.39$ and $+0.27$). Qualitative inspection reveals the mechanistic cause: unconstrained base models respond to culturally rich prompts with wandering, associative continuations filled with encyclopedic elaboration, boosting nuance and depth sub-scores. For concrete controls, however, base models quickly collapse into repetitive single-word definitions. In the instruction-tuned regime, where output length and formatting are calibrated, models readily produce concise, faithful definitions for standard controls, but struggle to compress culturally nuanced lexical gaps into English without sacrificing nuance or hallucinating commercial associations.

Confound regression. In an OLS regression controlling for character length, token count, language, and category, the untranslatability coefficient remains stable ($\beta = -0.012$, $p = 0.29$ for Llama-3-8B-Instruct; $\beta = +0.042$, $p = 0.063$ for Mistral-7B). This confirms that lexical gaps do not impose an unconditional, blanket suppression across all architectures; rather, the behavioral degradation is specific to instruction-tuned alignment.

Qualitative mechanisms in generated explanations. Table 4 illustrates the linguistic mechanisms governing model behavior:

- **Commercial co-occurrence distraction:** For Spanish *Sobremesa*, the model begins with the correct social definition ("after-dinner conversation") but immediately hallucinates a brand association ("Sobremesa is a wine that invites you to linger"), dropping its faithfulness score to 0.738.

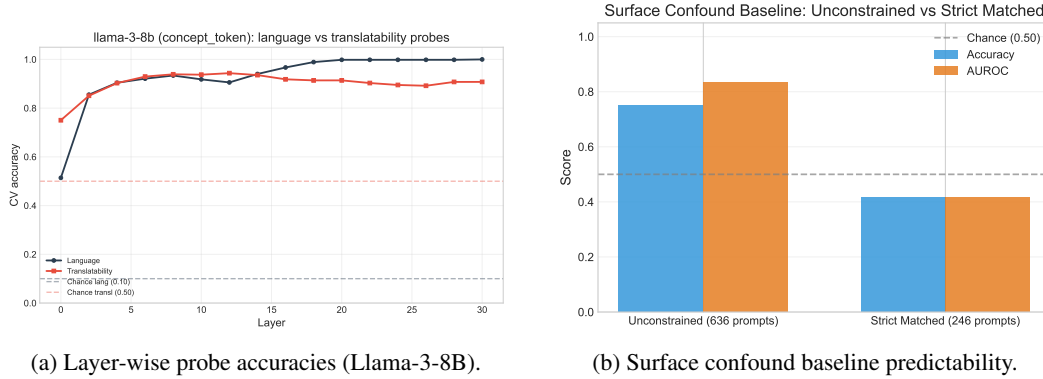


Figure 2: Probe representations and confound baselines. (a) Residual stream linear probes across layers; (b) Non-neural surface feature baseline predicting condition on unconstrained vs. strictly matched benchmark.

- **Semantic flattening:** For Japanese *Tsundoku*, the model accurately identifies book accumulation but flattens the cultural concept into mere “hoarding”, stripping the affectionate self-deprecation intrinsic to Japanese literary culture.
- **Loanword resilience:** For German *Schadenfreude* and *Wanderlust*, both established loanwords in English, the model achieves high faithfulness (0.967 and 0.904) due to dense pretraining exposure in English web corpora.

4.2 Probing representations: language dominance and confound baselines

The confound mirage in linear probing. As shown in Figure 2b, a simple logistic regression trained purely on non-neural surface confounds (character length, token count, language, category) predicts whether a concept is untranslatable with **75.0% accuracy** (AUROC = 0.834) on unconstrained data. On our strictly matched subset, this baseline collapses to **41.7% accuracy** (AUROC = 0.417), operating at chance level. This demonstrates why strict matching is mandatory: in unconstrained settings, high probe accuracy can merely reflect surface orthography rather than conceptual understanding.

Normalized representational geometry. Evaluating residual streams under GroupKFold reveals a clear geometric hierarchy:

- **Language dominance:** 10-way language probes achieve peak accuracy ≈ 1.000 across all models, corresponding to a **100.0% normalized improvement over chance** (NormAcc = 1.00). Language identity is organized as the primary linear manifold in multilingual residual streams.
- **Translatability subspace:** Binary translatability probes peak at 0.940–0.958 (NormAcc = 0.88–0.92), while within-language probes peak at 0.84–0.88 at the concept token (NormAcc = 0.68–0.76) and 0.92–0.94 at the last token (NormAcc = 0.84–0.88).

Residual-stream activations exceed the surface confound baseline by **+20 percentage points** on unconstrained data and by **+52 percentage points** on the strictly matched subset. Thus, while language identity is the dominant linear coordinate, residual streams encode an authentic translatability direction that survives strict confound control.

4.3 English routing: the diagnostic mirage of ERR

Figure 3 and Table 5 report the logit-lens analysis across all four architectures.

A definitive negative result. On strictly matched pairs, **differences in English probability mass are statistically indistinguishable from zero** across all four models ($|d| \leq 0.133$, all $p > 0.29$).

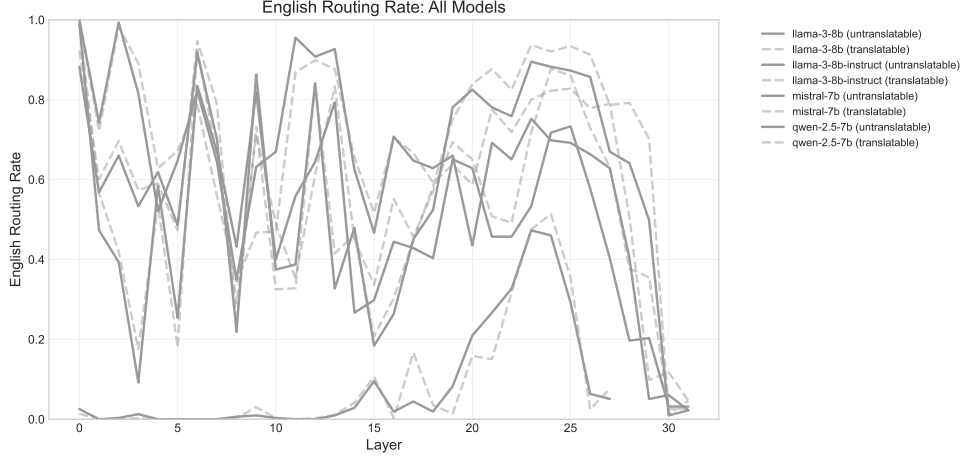


Figure 3: English Routing Rate across layers (untranslatable solid, translatable dashed).

Model	p_{eng} Untrans.	p_{eng} Transl.	Cohen’s d [95% CI]	t -statistic	p -value
Llama-3-8B	0.4161	0.4152	+0.016 [−0.23, +0.26]	+0.128	0.898
Llama-3-8B-Instruct	0.3949	0.3998	−0.071 [−0.32, +0.17]	−0.554	0.581
Mistral-7B-v0.3	0.5245	0.5351	−0.133 [−0.38, +0.11]	−1.040	0.300
Qwen2.5-7B	0.1864	0.1798	+0.092 [−0.15, +0.34]	+0.722	0.472

Table 5: English probability mass (p_{eng}) on strictly matched pairs. All effects are null ($p > 0.29$).

236 Top-1 ERR exhibits sharp early peaks in layers 1–6 (reaching 60–70% in Llama and Mistral) before
237 collapsing to near zero at output layers where tokens resolve into the target script.

238 **Why ERR fails as a cultural diagnostic.** Cross-model correlations reveal that early ERR peaks
239 track tokenizer vocabulary composition ($r = 0.72$ between Llama and Mistral; near zero vs. Qwen)
240 rather than cultural semantics. In early layers, un-embedded hidden states project onto the leading
241 singular vectors of W_U , which correspond to high-frequency Latin unigram tokens. Because Latin
242 tokens dominate transformer vocabularies ($> 70\%$ in Llama and Mistral), any uncalibrated projection
243 yields an artificial “English” peak. ERR measures vocabulary entropy and unigram priors, not
244 conceptual anglicization.

245 5 Discussion and Limitations

246 **Implications for multilingual interpretability.** Our empirical findings offer four methodological
247 lessons for the NLP community:

- 248 1. **The confound baseline imperative:** Probing studies must benchmark against non-neural
249 surface baselines. In linguistic tasks, word length and domain distribution can masquerade
250 as high probing accuracy.
- 251 2. **Condition-blind evaluation:** Behavioral evaluations of cultural competencies must en-
252 force condition blinding to prevent evaluators or heuristics from penalizing descriptive
253 circumlocution.
- 254 3. **The fragility of unembedding projections:** The logit lens reflects vocabulary-level singular
255 value structures in early layers; aggregate routing statistics like ERR should not be treated
256 as proxies for conceptual thought.
- 257 4. **Alignment as cultural compression:** Instruction tuning optimizes models for concise
258 conversational responses, which can inadvertently flatten nuanced cultural concepts that
259 require circumlocution.

Limitations. We highlight several limitations of our study:

- **The contested nature of untranslatability:** In linguistic theory, whether concepts are fundamentally untranslatable or merely lack monolexemic equivalents is debated [6, 1]. We operationalize untranslatability as a documented lexical gap in bilingual lexicography.
- **Sample size from strict matching:** Enforcing exact category and script matching reduces the benchmark to 41 pairs (82 concepts). While statistically powered for within-pair permutation tests, expanding coverage to indigenous and low-resource language families remains essential.
- **Annotation scope:** Reference glosses were derived from established lexicographic sources rather than native-speaker consensus panels. While condition blinding removes experimental bias, large-scale native-speaker validation is a vital next step.
- **Observational scope:** We analyze activations under greedy generation; we do not perform causal activation patching or steering.

Broader impacts. This work promotes methodological hygiene by demonstrating that unnormalized mechanistic metrics can produce misleading claims of linguistic bias. Transparent reporting of negative results prevents misdirection in mechanistic interpretability.

6 Conclusion

In a controlled lexical-gap inventory, models show lower cultural-explanation faithfulness in instruction-tuned models, while ERR does not provide a stable diagnostic. Measuring cultural representation in multilingual LLMs requires strict control matching, condition-blind behavioral evaluation, and representational probes rather than reliance on English routing proxies.

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1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: Claims are precisely calibrated: we report condition-blind generation faithfulness on matched pairs, probe performance against surface confound baselines, and establish that ERR is null on strictly matched controls.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 details four limitations: the contested nature of lexical gaps, sample size from strict matching, reliance on lexicographic reference glosses, and observational nature.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

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Justification: Sections 3.1–3.5 describe the Hungarian matching optimization, model parameters, condition-blind rubric formulas, and probe CV setup.

5. Open access to data and code

Question: Does the paper provide open access to the data and code?

Answer: [Yes]

Justification: Anonymized code, prompt sets, activation metadata, and scoring scripts are included with the submission repository.

6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [Yes]

Justification: Model IDs, decoding settings ($T = 0$, $\text{max_new_tokens} = 128$), GroupKFold splits, and regularization are fully documented.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined?

Answer: [Yes]

Justification: All tests report exact p -values, paired permutation tests (10,000 permutations), and Cohen’s d with analytical 95% confidence intervals (Tables 3, 5).

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [Yes]

Justification: Models were evaluated using open-weight checkpoints on NVIDIA L4 GPUs with execution times under 4 GPU-hours.

9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

353 Answer: [Yes]
 354 Justification: The research analyzes open linguistic concepts from public dictionaries without
 355 human subjects or dual-use risk.

356 **10. Broader impacts**
 357 Question: Does the paper discuss potential positive societal impacts and negative societal
 358 impacts?
 359 Answer: [Yes]
 360 Justification: Section 5 discusses the importance of objective metric validation and avoiding
 361 cultural stereotyping from unnormalized model metrics.

362 **11. Safeguards**
 363 Question: Does the paper describe safeguards for responsible release?
 364 Answer: [N/A]
 365 Justification: The study releases only small linguistic probe sets.

366 **12. Licenses for existing assets**
 367 Question: Are creators of existing assets properly credited?
 368 Answer: [Yes]
 369 Justification: Model origins (Meta, Mistral, Qwen) and citations are cited.

370 **13. New assets**
 371 Question: Are new assets well documented?
 372 Answer: [Yes]
 373 Justification: The 41 strictly matched pairs and diagnostics are documented in Section 3.1
 374 and Table 1.

375 **14. Crowdsourcing and research with human subjects**
 376 Question: Does the paper include human subjects instructions?
 377 Answer: [N/A]
 378 Justification: No human subjects experiments were conducted.

379 **15. Institutional review board (IRB) approvals**
 380 Question: Were IRB approvals obtained?
 381 Answer: [N/A]
 382 Justification: No human subjects or private personal data were involved.

383 **16. Declaration of LLM usage**
 384 Question: Does the paper describe the usage of LLMs in research?
 385 Answer: [Yes]
 386 Justification: LLMs are the subject of empirical investigation; LLM-as-judge is disclosed as
 387 a secondary analysis interface in Section 3.3.